

papers published, patents and corporate investment in AI,” he says. “India has software services giants such as TCS (Tata Consultancy Services) and Infosys which serve almost all Fortune 500 companies. They can build infrastructure and applications around Generative AI for these clients and dramatically lift their engineers’ productivity using AI,” he adds.

India arriving late at a high-tech party is not new. It was late for the Cloud revolution in computing, was initially tentative in the space race and has often erred on the side of caution when others dared to bet big. And yet, there’s one thing the country has never lacked: the resolve to play catch-up with—and sometimes leapfrog—rivals. With AI rapidly becoming the backbone of digital economies, the Indian government has set itself the goal of building a \$1 trillion AI industry by 2030.

But winning in foundational AI—a form of generative AI models trained on vast amounts of diverse data, enabling them to be adapted for a wide range of applications—demands more than ambition. From graphic processing units (GPUs) and high-bandwidth memory to advanced chip fabrication units and specialised electronic design automation (EDA) tools, the tech stack underpinning AI is expensive and complex.

Consider the scale of investment: the US has poured about \$10 billion into its AI ecosystem and China has invested between \$15 billion and \$30 billion. India’s AI investment is just \$1.2 billion.

Morgan Stanley estimates that China’s AI investments could break even by 2028 and deliver a 52% return on invested capital by 2030, a projection that underscores the economic stakes of foundational AI.

A debate is under way, however,

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NITIN MITTAL
Deloitte Global
AI Leader



“India has all the ingredients to lead: diversity, digital scale, and deep talent. What we need now is the will.”

DR VIVEK RAGHAVAN
Co-founder,
Sarovam AI

on whether investment alone guarantees success. Cost-effective launches of smaller AI models are proving that innovative thinking may sometimes matter as much as deep pockets.

Even so, the scale of the challenge confronting India is formidable.

MORE THAN CATCHING UP

Without building domestic capacity, India will run into “structural ceilings,” says Nitin Mittal, Global AI Leader at Deloitte. “In the short run—from 2025 to 2030—pooling demand, like MeitY’s bulk GPU procurement, and leveraging India’s design talent can drive growth,” says Mittal.

“But beyond 2030, without indigenous advanced-node fabs, high bandwidth memory (HBM) and 2.5-D/3-D packaging, India risks forfeiting potential AI-led GDP (gross domestic product) gains, ceding standards-setting power, and seeing its top talent migrate to countries that control the Silicon stack,” says Mittal. MeitY is short for the Ministry of Electronics and Information Technology. 2.5-D and 3-D are advanced semiconductor packaging technologies.

Under the IndiaAI Mission, the government has procured around 34,000 GPUs from giants like Nvidia Corp., Advanced Micro Devices Inc. Amazon Web Services Inc. and Intel Corp. This is just a fraction of what it needs.

The world is, meanwhile, racing ahead. Companies like Perplexity, the AI-powered search engine, are redefining the browser experience. Western enterprises are weaving AI into the fabric of business processes. Global attention is rapidly shifting towards the next frontier: Artificial General Intelligence (AGI) that possesses human-level cognitive abilities.

India has already missed the first wave of large language models (LLMs), as AI models from the US, China, France, and even the Unit-